

Phase 5 Report: Integration, Security and Compliance Testing

Residential House Rent Control and Administration System - Gate G5 Package

Legal Basis: Proclamation 1320/2016 + Directive 7/2016 + Model Agreement

Contents: Test Plan, Level Reports, Security Assessment, Compliance Matrix

Baseline: SRS v1.1 (CR-01 Trilingual: Amharic, English, Afan Oromo)

Gate: G5 - Accept Test Evidence and Authorize UAT

Date: September 2026

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1. Introduction and Phase 5 Scope

Phase 5 is the proving ground that the implementation plan reserved for the assembled system: integration, security and compliance testing. The plan's objective for this phase reads that the platform must be proven against integrations, load, attack, and above all the law. In line with that objective, the phase exercised the platform at four levels simultaneously. Integration testing bound the payment ledger to a bank sandbox and party onboarding to an identification service, including the failure and timeout behaviour the plan explicitly requires. System and regression testing re-ran the entire automated battery built during construction. Security testing implemented and verified the two controls the non-functional requirements call out by name: access control between tiers and audit-trail integrity. Finally, the legal compliance test matrix, the project's signature quality instrument, was executed in full, so that every traced article of the three source documents is now exercised by at least one named test case with its result recorded.

The phase also carried responsibilities forward from earlier gates. The owner approved Gate G4, releasing this phase with four open items travelling alongside: O1 (the directive's fine-ladder figures pending official text), O-7 (woreda register reconciliation), O-8 (Afan Oromo legal glossary certification under CR-01) and O-9 (the committee hearing window parameter). Each of these received an explicit disposition in this phase, recorded in chapter 8; O-9 is now closed and the other three carry documented dispositions that do not block UAT entry. The change request CR-01 remained fully in force: the trilingual commitment is itself a tested compliance row, and the testing console tab renders the same three languages as the rest of the platform.

This report is the Gate G5 package. Its chapters follow the deliverables the plan names for this phase: the test plan (chapter 2), the level-wise test reports (chapters 3, 4 and 6), the security assessment (chapter 5), and the compliance matrix with results attached (chapter 7). All evidence is generated by persisted scripts and regenerable on demand: the automated battery runs through scripts/quality/run-all.ts, the load profiles through scripts/load/perf.ts, and the console exposes the same evidence live through the new P5 tab. At the time of writing the battery stands at 113 passing test cases with 0 failures, and the compliance matrix shows 47 of 47 rows passing with none failed or incomplete.

2. Test Plan and Method

The test plan follows the V-model commitment of the implementation plan: every requirement maps to at least one test case, and every legal constraint maps to a named compliance case. Five levels run in this phase. Unit and seed-config suites, built during

Phases 3 and 4, continue as the regression floor. The integration suite is new in this phase and exercises the external service adapters. The security suite is likewise new and tests the authorization guard and the audit chain directly at the enforcement point the API routes use. The legal compliance suite re-expresses the SRS traceability matrix as executable cases. Finally, the performance suite measures the running platform under load rather than in isolation. All five levels run automatically through one command, and the same command attaches per-case results to the compliance matrix rows, which is how the matrix stays a living artifact rather than a one-off document.

Table 1: Test levels, suites and case counts

Level	Suite	Cases	What it proves
Regression floor	tests/seed-config.test.ts + tests/phase4-domain.test.ts	51	Seed configuration integrity (hierarchy, catalogues, trilingual resources) and the legal rule engine's date, calendar, ceiling and penalty mathematics.
Integration	tests/integration.test.ts	6	Bank sandbox settlement before ledger writes; decline and timeout behaviour; identity match, mismatch and deferred-outage behaviour.
Security	tests/security.test.ts	13	Tier-based authorization (403 paths), separation of duties, capability-matrix integrity, audit chain append, tamper detection.
Legal compliance	tests/compliance-legal.test.ts	43	Every binding article of the Proclamation, the Directive and the model agreement exercised through the service layer against the live database.
Performance	scripts/load/perf.ts profiles	3 profiles	Woreda steady state, city peak and registration end-to-end latency against the NFR-01 targets.

The test environment is the Phase 3 environment inventory itself: the developer database seeded from the official configuration (131 organization units, 13 roles, 5 identification

types, 3 languages, 23 localization resources, 14 penalty parameters, the model contract and the June calendar), behind the same HTTP API the console uses. Test data is created by the suites themselves through the public service layer, and the demonstration dataset is restored by reseeding plus the 33-check end-to-end walkthrough after test runs. Workload figures for performance testing are stated as explicit assumptions in chapter 6, where the sandbox profile is scaled to the container and projected linearly to the full-city NFR-01 target of 500 concurrent office users; the projection method and its margin are documented there rather than silently assumed. Defect severity follows the plan's definitions: severity one blocks a legal gate or corrupts a record, severity two breaks a golden path with a workaround, severity three is a cosmetic or convenience defect. Entry criteria for this phase were the Gate G4 approval and the closure of all in-phase severity-one and two defects; exit criteria are stated at the end of chapter 10 for the owner's decision.

3. Integration Test Report: Payment and Identification Sandboxes

Open item O3 recorded during requirements analysis that the production banking and e-payment partner APIs were unknown pending procurement, and the plan answered by requiring the platform to integrate against sandboxes with explicitly modelled failure behaviour. Phase 5 delivered exactly that: two adapters, a bank gateway and an identification service, each behind a narrow interface that the production partner will implement at procurement without touching ledger or registration code. The payment adapter settles an electronic transfer before the ledger entry is written, and returns a provider reference that is stored on the receipt. The identification adapter answers match, mismatch or timeout for the document types the directive accepts, and the party workflow treats a timeout as a deferral, never as a blocker, so that a worda office does not stop working because a partner service is down.

Table 2: Integration failure and timeout drills (all executed as named cases)

Drill	Stimulus	Required behaviour	Result
TC-INT-01	Electronic rent on a registered contract	Gateway settles first; provider reference stored on the ledger entry.	Pass
TC-INT-02	Provider declines the transfer (forced DECLINE mode)	HTTP 424 with the violated rule surfaced; no ledger entry written; count verified unchanged.	Pass

Drill	Stimulus	Required behaviour	Result
TC-INT-03	Provider never answers (forced TIMEOUT mode)	HTTP 504; no ledger entry; the same reference settles cleanly on retry.	Pass
TC-INT-04	Identity service reports no record for the document	Party created PENDING with the mismatch note recorded for manual follow-up; no silent verification.	Pass
TC-INT-05	Identity service outage (forced TIMEOUT mode)	Party created with DEFERRED note; the office workflow continues; registrar verification still possible.	Pass

Two findings worth recording came out of this level. First, the decline and timeout drills proved the ordering invariant that the law actually needs: because settlement precedes the database write, the ledger can never diverge from the bank, which is the admissible-evidence guarantee of Proclamation Article 14 expressed mechanically. Second, the identity timeout drill validated the graceful-degradation rule as a product decision: a deferral keeps the office open while still refusing to auto-verify an unverified identity, preserving the registrar's statutory role in verification. The production bindings remain open item O3 by design; the interfaces, the failure taxonomy and the ledger invariants are now fixed and tested, so procurement changes configuration rather than code.

4. System and Regression Test Report

The automated battery stands at 113 passing cases across five suites with 0 failures, re-runnable with one command and executed after every change in this phase. Beyond the unit and integration levels, the end-to-end walkthrough re-ran the full golden path over the live HTTP API: party onboarding with verification, property registration with its exemption clock, the complete registration lifecycle from presentation through the nine-point checklist, certification, stamping and registration into the registry book, the payment ledger with its cash-referral rule, the adjustment lifecycle from draft through June 1 publication to June 30 effect, increase validation against the published ceiling, the complaint pipeline with its statutory clocks, the committee appeal, the penalty ladder with the vacancy surcharge, replication, backup, aggregation and publication. The walkthrough passed all 33 checks after the security layer went live, which is itself a meaningful regression result: the golden path still works end to end when every mutating call must now carry an authorized officer's identity.

Table 3: Suite results at the time of reporting

Suite	Cases	Status	Notable coverage
Seed configuration (Phase 3)	17	Pass	Official structure (11 sub-cities, 118 woredas), trilingual catalogues, model contract sections, environments.
Legal rule engine (Phase 4)	34	Pass	Working-day calendar, exemption clocks, lease validation, ceilings, penalty bands, fine caps, book numbering.
Integration (Phase 5)	6	Pass	Settle-before-ledger, decline/timeout drills, identity mismatch and deferral.
Security (Phase 5)	13	Pass	403 paths per capability, separation of duties, chain append and tamper detection.
Legal compliance (Phase 5)	43	Pass	Named cases TC-P/TC-D/TC-MA/TC-N covering all 47 matrix rows.

Regression discipline held during the phase: three defects were found and fixed in-phase with the battery re-run green afterwards (chapter 9 records them). The suites are deliberately written against the service layer and the HTTP guard rather than against database internals, so they remain valid when infrastructure changes; the compliance suite in particular creates its own fixtures through public functions, which keeps it executable against any seeded environment, including the pilot configuration when it arrives.

5. Security Assessment (OWASP ASVS Level 2 Focus Controls)

NFR-04 sets OWASP ASVS Level 2 as the security baseline, and the plan's Phase 5 names two controls for explicit verification: access control between tiers and audit-trail integrity. Both controls were implemented at the enforcement point during this phase rather than assessed as aspirations. Every mutating API route now names a capability, and the capability names the role codes allowed to execute it; twenty-six capabilities cover the full route surface, mapped directly to the directive's tier duties. The acting officer is resolved from the request against the staff register, anonymous calls are refused, wrong-tier roles are refused with the required roles named in the denial, and the stamping desk is a distinct capability from certification exactly as Directive Article 9 describes. Successful actions

append to an append-only audit chain in which every event hashes its predecessor with SHA-256 over the full payload, so any alteration of history invalidates every later hash.

Table 4: Assessment results by control area

Control	ASVS reference	Verification	Result
Tier-based authorization on all mutating routes	V4.1	TC-SEC-01..05: anonymous 403, wrong-tier 403, separation of duties, positive paths per role, matrix mirrors directive tiers.	Pass
Ministry-only publication	V4.1	TC-P18 positive and negative across five other roles.	Pass
Immutable audit trail	V7.2-7.3	TC-SEC-06..08: append on every state change, full-chain recompute intact, mid-chain tamper detected and located.	Pass
Audit content	V7.1	TC-SEC-08b: actor, role, org, action, entity, references and 64-character hashes on every probe event.	Pass
Integration error mapping	V4 / V14	Decline and timeout mapped to 424/504 with no partial writes (chapter 3).	Pass
Input handling	V5	All persistence flows through parameterized ORM queries; unknown enum values rejected by the service layer with the legal rule named.	Pass

One finding is recorded honestly and with a disposition. Read endpoints in the demonstration console are intentionally open, because the console is the owner's review surface; in a production deployment, session-based read authorization must be enforced alongside the mutating-route guard, and sensitive reads (party identity data) must be logged per NFR-07. This is finding F-1, classified severity three for the pilot scope because no state change is possible without an authorized officer, with its closure scheduled in the go-live hardening of Phase 8 where real session management binds to the national identity provider. The assessment also notes the residual trust boundary of the demo staff codes: they exist to demonstrate role separation in review; production authentication replaces

them at the same integration point, and the capability matrix they exercise is the same matrix the tests verify.

6. Performance Test Report

NFR-01 sets the acceptance targets: interactive screens within three seconds at the 95th percentile under a full-city load of 500 concurrent office users and 5,000 public portal users, and registration submission within five seconds end to end. The load harness drives the running platform through the same HTTP API the console uses, with a weighted read mix representing a woreda office day (files, parties, payments, complaints, deadlines) plus public portal pressure on the publication feed and the heavier boot aggregation. Three profiles run: a woreda steady state, a city-wide peak, and a sequential registration-path measurement. The measured results at the sandbox scale are shown below; all profiles ran with a zero percent error rate after the integration regression fix, which is itself a stability indicator under sustained concurrency.

Table 5: Measured profiles versus NFR-01 targets

Profile	Concurre ncy	Requests	p50 (ms)	p95 (ms)	p99 (ms)	RPS	Errors
WORED A	25	2054	168	358	521	135.9	0%
CITY_PE AK	60	3039	357	672	792	149.8	0%
REG_E2E	20	-	43	64	-	-	0%

The targets are met with wide margins at the measured scale: the interactive p95 of 672 ms at city-peak concurrency sits at roughly a fifth of the three-second target, and the registration path's p95 of 64 ms is far inside the five-second end-to-end budget. The projection to the full NFR-01 load is stated explicitly rather than implied: the sandbox executes at roughly one-eighth of the 500-office concurrency, and the latency curve observed between the woreda and city-peak profiles rises sub-linearly (p95 grows by about 88 percent when concurrency grows by 140 percent), which indicates headroom rather than saturation at this scale. Three engineering caveats accompany the projection honestly: the sandbox database is SQLite in the developer environment while production will run the same schema on a production database engine; the p99 tail includes cold-route compilation in the development server that production builds do not have; and public portal pressure was simulated at a 2:1 read ratio rather than the 10:1 ratio of the live target, with the heavy portal pages being exactly the publication reads measured here. The Phase 8 go-live checklist therefore carries a re-run of these profiles in the staging environment as a standing item, using the same persisted harness with raised concurrency.

7. Legal Compliance Matrix with Results

This chapter is the deliverable the plan calls the project's signature instrument: every traced article paired with the system behaviour that proves it, and every rule tested at least once by a named case whose result is attached. The matrix below is generated directly from the platform's matrix definition and the latest battery run, so what the owner reads here is what the machine executed. Verification kinds distinguish fully automated rules, parametric rules where the law's figures live as configurable seed parameters (asserted for presence, structure and legal basis), and documentary duties outside software behaviour where the platform records the audit surface. The G5 exit criterion of the plan is that the compliance matrix shows no failed legal rule; the matrix currently shows 47 of 47 rows passing with 0 failed and 0 incomplete.

Table 6: Compliance matrix — 47/47 rows passing

ID	Source	Rule	Module	Kind	Result
TC-P02	Proc. Art. 2(2-4)	Residential property status classified as new construction, previously-rented vacant, or occupied; drives the exemption clock.	M2	AUTOMATED	PASS
TC-P03	Proc. Art. 3(3)	Houses administered by federal/city organs are outside this proclamation's rent-ceiling regime; scope recorded in city configuration.	M13	PARAMETRIC	PASS

ID	Source	Rule	Module	Kind	Result
TC-P04	Proc. Art. 4	Rental of residential houses requires a written, certified and registered contract; unregistered contracts have no legal effect; payments attach only to registered files.	M4	AUTOMATED	PASS
TC-P05	Proc. Art. 5; Dir. Art. 4	The Bureau publishes the model rental agreement; the studio amends it under version control without touching superseded versions.	M3	AUTOMATED	PASS
TC-P06	Proc. Art. 6	Minimum lease term of two years; shorter terms are refused at intake.	M4	AUTOMATED	PASS
TC-P07	Proc. Art. 7; Dir. Art. 8(2)	Legacy (pre-proclamation) contracts are registered during the 30+3 day campaign with a legacy annotation distinguishing migrated records.	M4	AUTOMATED	PASS

ID	Source	Rule	Module	Kind	Result
TC-P08	Proc. Art. 8; Dir. Art. 11	Annual rent adjustment studied, published June 1, effective June 30; increases beyond the ceiling refused.	M5	AUTOMATE D	PASS
TC-P09	Proc. Art. 9	First-time pricing of new (never-rented) houses is free; the adjustment engine skips houses inside the exemption window.	M5	AUTOMATE D	PASS
TC-P10	Proc. Art. 10	Exemption clocks: 48 months for new construction, 24 months for previously-rented vacant houses; vacancy surcharge bands 5-25 percent by vacancy years.	M2/M9	AUTOMATE D	PASS
TC-P12	Proc. Art. 12	Advance payment capped at two months' rent.	M6	AUTOMATE D	PASS

ID	Source	Rule	Module	Kind	Result
TC-P13	Proc. Art. 13; Dir. Art. 22	Rent is payable only through bank or legal electronic channels; cash payments auto-open the 10 percent referral case; electronic transfers settle at the gateway before any ledger entry.	M6	AUTOMATED	PASS
TC-P14	Proc. Art. 14	In any proceeding the registered contract and electronic payment records are the admissible evidence; the ledger therefore refuses entries outside registered contracts.	M6/M9	AUTOMATED	PASS
TC-P15	Proc. Arts. 15-17	Termination regime including no-notice grounds; termination without statutory notice is a sanctionable offense (3 months' rent).	M4/M9	PARAMETRIC	PASS

ID	Source	Rule	Module	Kind	Result
TC-P18	Proc. Art. 18(3)	National statistics and ceiling rates are published for the public; publication is a Ministry-role capability.	M11/M12	AUTOMATED	PASS
TC-P19	Proc. Art. 19(5); Dir. Arts. 14(4), 16(6)	The regulatory body establishes the IT system that registers contracts, records payments and produces statistics; this platform is that system and boots from the seeded hierarchy.	M13	AUTOMATED	PASS
TC-P20	Proc. Art. 20; Dir. Arts. 17-18	Tenant complaints (overcharge, illegal eviction, etc.) are received through multiple channels and registered against the eight statutory grounds.	M8	AUTOMATED	PASS
TC-P22	Proc. Arts. 20-23	The complaint decision is due within 30 working days of receipt; the decision clock opens at intake and closes at decision.	M8/M12	AUTOMATED	PASS

ID	Source	Rule	Module	Kind	Result
TC-P24	Proc. Art. 24	Appeals to the hearing committee must be filed within 15 days of the decision; late appeals are refused.	M8	AUTOMATED	PASS
TC-P25	Proc. Arts. 25-26	The committee hears appeals within the hearing window and may escalate to the courts; window encoded as a confirmed parameter (O-9 verified 30 days). — O-9 closed: committee window = 30 days, basis cited in parameter seed	M8	AUTOMATED	PASS
TC-P29	Proc. Arts. 29-32	Administrative fines are capped at three months' rent of the affected contract; the cap applies at computation time.	M9	AUTOMATED	PASS
TC-P29R	Proc. Art. 29	Regional implementation is by directive; per-city parameter sets exist so Addis Ababa values never hard-code.	M13	PARAMETRIC	PASS

ID	Source	Rule	Module	Kind	Result
TC-D04	Dir. Art. 4	Contract content follows the annexed model; the seeded model carries the required section structure in trilingual renderings.	M3	AUTOMATED	PASS
TC-D05	Dir. Art. 5	Registration requires the identification documents (five accepted types), ownership evidence and proxy documents with two witnesses.	M1/M2	AUTOMATED	PASS
TC-D06	Dir. Art. 6	Registration happens in person at the woreda where the house is located; files are opened on a woreda-tier org unit with a computer-typed form.	M4/M13	AUTOMATED	PASS
TC-D07	Dir. Art. 7	The nine-point document verification checklist must be completed before certification.	M4	AUTOMATED	PASS

ID	Source	Rule	Module	Kind	Result
TC-D08	Dir. Art. 8	Registrar certification acts follow checklist completion; the registry records who certified.	M4	AUTOMATE D	PASS
TC-D09	Dir. Art. 9	Stamping desk applies the office round stamp; database entry and the uniform registry book follow with sequential numbering (50 entries per page).	M4	AUTOMATE D	PASS
TC-D10	Dir. Art. 10	Amendments must be registered within the 30-day window; rate-affected amendments checked against the incoming rate set.	M4/M5	AUTOMATE D	PASS
TC-D11	Dir. Art. 11	Annual increase study feeds publication June 1 and effect June 30; the calendar engine enumerates the statutory events.	M5/M13	AUTOMATE D	PASS

ID	Source	Rule	Module	Kind	Result
TC-D12	Dir. Art. 12	Registration corrections are recorded as annotations on the file, never as silent rewrites of certified data.	M4	AUTOMATED	PASS
TC-D13	Dir. Art. 13	Hard and soft records, document scanning, upward tier replication (hop counts by tier) and per-tier nightly backups.	M10	AUTOMATED	PASS
TC-D14	Dir. Arts. 14-16	Bureau, sub-city and woreda duties map to the role catalogue with tier scopes; data aggregates upward through the hierarchy.	M13/M12	AUTOMATED	PASS
TC-D17	Dir. Art. 17	Exactly eight complaint grounds are configured, matching the directive's list.	M8	AUTOMATED	PASS
TC-D18	Dir. Art. 18	Complaints arrive via walk-in, phone, web, letter and third-party referral; verbal statements require a minimum factual description.	M8	AUTOMATED	PASS

ID	Source	Rule	Module	Kind	Result
TC-D19	Dir. Art. 19	Admissibility verification, investigation, decision and notice proceed in statutory order with deadline escalation.	M8	AUTOMATED	PASS
TC-D20	Dir. Art. 20	Control teams act on sub-city authority, show identification credentials on visit, and monitor vacancy.	M7	AUTOMATED	PASS
TC-D21	Dir. Art. 21	Prohibited acts (demanding excess advance, unregistered renting, unauthorized increase, noticeless termination, illegal eviction) map one-to-one to offense parameters. — O1 tracked: offense-level figures configurable as PENDING_OFFICIAL_TEXT pending official Amharic text	M9	PARAMETRIC	PASS

ID	Source	Rule	Module	Kind	Result
TC-D22	Dir. Art. 22	Penalty ladder: cash-payment 10 percent referral rule, vacancy surcharge bands 5/10/15/20/25 percent, tax escalation and court referral; cap enforced.	M9	AUTOMATED	PASS
TC-MA1	MA cl. 1-3	Lessor, lessee and property blocks are populated from registered party and property profiles; they cannot diverge from the registry.	M3/M4	AUTOMATED	PASS
TC-MA4	MA cl. 4(1-4)	House status selector drives the exemption clock; rent in words and figures; term at least two years; service-charge payer enum.	M4	AUTOMATED	PASS
TC-MA6	MA cl. 6(1)	Advance and monthly due day captured; advance capped at two months (Proc. Art. 12).	M6	AUTOMATED	PASS

ID	Source	Rule	Module	Kind	Result
TC-MA8	MA cl. 8; signatures + 3 witnesses	The contract takes effect only upon regulator certification; execution requires three witnesses with identification.	M4	AUTOMATE D	PASS
TC-N04	NFR-04 (OWASP ASVS L2, V4)	Access control between tiers: worded, sub-city, Bureau, committee, Ministry and system roles can execute only their own capabilities; anonymous calls are refused.	Platform	AUTOMATE D	PASS
TC-N07	NFR-07 (ASVS V7)	Every state-changing action appends an immutable audit event; the SHA-256 chain detects any alteration of history.	Platform	AUTOMATE D	PASS
TC-INT01	Plan §5.6; O3	Payment and identification sandboxes integrate with defined failure and timeout behaviour; identity timeouts degrade gracefully with a deferral note instead of blocking the office.	M1/M6	AUTOMATE D	PASS

ID	Source	Rule	Module	Kind	Result
TC-N06	NFR-06 / CR-01	Amharic, English and Afan Oromo interfaces from seeded localization resources; certified Afan Oromo legal glossary tracked as open item O-8 with fallback. — O-8 tracked: Afan Oromo legal glossary certification pending; fallback per NFR-06	M13	PARAMETRI C	PASS
TC-N12	NFR-12	Sequential register numbering enforced database-side; certified contract records immutable after certification (corrections via annotations only).	M4	AUTOMATE D	PASS

Two properties of this matrix deserve the owner's attention beyond the pass column. First, coverage is clause-level where the documents bind behaviour: the Proclamation rows run from property classification through contracts, ceilings, exemptions, payments, evidence, disputes, committee and court escalation to the fine cap and regional directives; the Directive rows run from the model contract's content through the nine-point checklist, certification, stamping, registers, replication, roles, complaint grounds and pipeline, control teams and the penalty apparatus; the model-agreement rows verify that the digital contract blocks derive from registered profiles and that certification is what gives the document legal effect. Second, the matrix is regenerable at every gate as NFR-11 requires: adding a legal rule means adding a matrix row and its named case, and the next battery run updates this chapter's evidence automatically.

8. Open Item Dispositions at Gate G5

The plan requires that open items be dispositioned against official texts and registers rather than silently inherited. All four items carried into this phase were dispositioned; one is closed and three carry explicit, non-blocking dispositions into the acceptance phase.

Table 7: Open item register after Phase 5

Item	Title	Disposition at Gate G5	Status
O1	Directive Art. 22 fine ladder exact figures (scanned two-column ambiguity)	Encoded as configurable PEN-* parameters with status PENDING_OFFICIAL_TEXT; penalty engine reads parameters, never constants. Confirmed figures replace seed values without code change. Carried into UAT legal review (Phase 6).	CARRIED
O-7	Woreda register reconciliation before pilot (provisional counts for some sub-cities)	131 org units seeded with per-unit CONFIRMED / PENDING_OFFICIAL_REGISTER basis badges; validator battery blocks promotion if the official total (118) drifts. Reconciliation against the official register remains a Phase 7 pilot prerequisite.	CARRIED
O-8	Afan Oromo legal glossary certification (CR-01)	Trilingual UI/notifications/documents ship with fallback per NFR-06; terminology marked PENDING_CERTIFICATION where legal phrasing needs the certified glossary. Certification workshop scheduled with the Bureau in Phase 6.	CARRIED

Item	Title	Disposition at Gate G5	Status
O-9	Committee hearing window parameter (Proc. Arts. 25-26)	Closed at Phase 5: hearing window = 30 days (committeeHearingDue), basis cited in the compliance matrix row TC-P25 and the parameter seed.	CLOSED

The O-9 closure deserves a sentence of explanation because it exemplifies how the platform treats legal parameters. The committee hearing window was encoded as a thirty-day configuration parameter during construction pending confirmation against the Proclamation's appeals articles; the compliance row TC-P25 now asserts the window arithmetically through the committee scheduling function with the article cited, and the parameter's basis string names the articles, so the legal review in the next phase can confirm it by reading one row and one function rather than re-deriving the analysis. The three carried items remain visible in the console's evidence tab with the same dispositions recorded here, and none of them blocks authorizing UAT: O1 and O-9-class figures are configuration, O-7 is a pilot-entry prerequisite of Phase 7, and O-8 is a certification workshop scheduled with the Bureau alongside the UAT legal validation.

9. Defect Log and In-Phase Fixes

Four defects surfaced during the phase and all were fixed before reporting, with the full battery re-run green afterwards. They are recorded here in the same honest register the project has kept since Phase 3, because a testing phase that reports no findings is not testing hard enough.

Table 8: Defects found and closed in Phase 5

#	Finding	How caught	Fix and verification
D-1	Audit hash payload omitted the event summary, so an edit limited to the summary field would not break the chain.	Security suite TC-SEC-07 tamper drill failed to detect the forced alteration.	Hash formula extended to the full payload; chain reset and re-verified; tamper drill now detects and locates the alteration.
D-2	Publications GET returned a server error after the guard refactor removed a shared helper from its imports.	Performance harness error-rate breakdown isolated the failing endpoint (14 percent of mixed traffic).	Import restored; endpoint verified; full load profile re-run at zero percent errors.

#	Finding	How caught	Fix and verification
D-3	End-to-end walkthrough calls did not carry officer identities, so the golden path collided with the new authorization layer.	E2E run failed with 403 on the first privileged act after RBAC went live.	Walkthrough updated to act as the correct officers per capability; all 33 checks pass under enforcement.
D-4	Compliance-suite fixtures captured the model contract id once, so a versioning probe that supersedes the active version stranded later cases.	Suite-internal failures with the active-version rule after the probe ran.	Fixture refreshed after versioning; suites re-runnable in any order with a fresh seed.

Severity classification: D-1 is severity two as a security-correctness gap in a control under assessment, closed in-phase; D-2 and D-3 are severity two regressions caught by the new evidence surfaces this phase exists to build; D-4 is severity three test-harness hygiene. No severity-one defect occurred. The defect log's most valuable output is structural: three of the four findings were surfaced by the evidence machinery itself (the tamper drill, the load-harness error breakdown, the RBAC-enforced walkthrough), which is the intended long-term behaviour of the platform's quality system.

10. Conclusion and Gate G5 Decision Request

The plan's exit criteria for this phase are met on the evidence of the preceding chapters. All critical tests pass: the automated battery stands at 113 passing cases with zero failures across regression, integration, security and compliance suites, and the 33-check end-to-end walkthrough passes under full authorization enforcement. The compliance matrix shows no failed legal rule: all 47 rows pass, covering the Proclamation's binding provisions, the Directive's operational articles and the model agreement's effect rules, with results attached by the same run that produced this report. The security assessment implements and verifies the two named controls, with one honest finding disposed to the go-live hardening phase. Performance targets are met with wide margins at the measured scale, and the projection to the full-city load is documented with its caveats and a standing re-run item for staging. The open items are dispositioned explicitly: O-9 closed, O1, O-7 and O-8 carried with non-blocking dispositions.

The requested decision is to accept the test evidence and authorize the start of Phase 6, User Acceptance and Legal Validation, in which the real roles judge the system: registrars run real registration files through the checklist, Bureau analysts run the adjustment publication flow, complaint officers run intake to decision, committee members hear a rehearsed appeal, and the Bureau's legal experts walk the compliance matrix article by

article, recording every deviation and its resolution. The platform, its evidence machinery and its open-item register are ready for that judgement.

Table 9: Gate G5 approval record

Role	Name	Signature	Date
Project Owner / Product Owner			
Regulatory Body Business Focal			
Quality / Test Lead			
Project Manager			

Table 10: Document version history

Version	Date	Author	Change Summary
1.0	2026-09-10	Quality & Test Team	Issued as the Gate G5 package: test plan, level-wise reports, security assessment, compliance matrix with results, open item dispositions, defect log, owner decision request.